

Yashi Huang

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ACADEMIC PROFILE

Ph.D. candidate in Operations Management with research and industry experience at the intersection of artificial intelligence, healthcare operations, business analytics, and decision sciences. Research combines machine learning, probabilistic forecasting, simulation, and optimization with real healthcare applications at a large U.S. cancer institute.

EDUCATION

- **University of Rochester** Sep 2021 – Dec 2026 (Expected)
Ph.D. Candidate, Operations Management
◦ GPA: 3.9/4.0
Rochester, NY
- **Columbia University** Sep 2018 – Feb 2020
M.S., Operations Research
◦ GPA: 3.9/4.0
New York, NY
- **Wuhan University** Sep 2014 – Jun 2018
B.A., Financial Engineering
◦ GPA: 3.8/4.0
Wuhan, China

TEACHING EXPERIENCE

- **Lab Instructor, OMG 402 (MSM 512) – Operations Management** Spring 2025
University of Rochester
◦ Supported student learning in core operations management topics and quantitative decision-making, connecting analytical methods to applied business problems.
◦ Overall instructor rating: 4.5/5.0.
- **Lab Instructor, GBA 465 – Introduction to Python Programming** Spring 2024
University of Rochester
◦ Led two hands-on lab sections in introductory Python programming, guiding students through coding exercises, debugging, and applications of Python to business analytics.
◦ Overall instructor rating: 4.4/5.0.

TEACHING INTERESTS

AI in Healthcare; Machine Learning and AI for Business; Healthcare Operations; Data Management and Analytics for Decision Making; Operations Management; Introduction to Python Programming.

RESEARCH EXPERIENCE

- **Dynamic Interval Prediction of Waiting Times in a Cancer Care Facility** Jul 2023 – Present
◦ Predicts a patient's wait as an interval rather than a point. Intervals are built from distributional models (random survival forests, neural networks, and a recurrent marked temporal point process that learns from the full event history) using adaptive quantile selection with conformal correction, and from point models using residual distribution.
- **Probabilistic Forecasting of Intraday Occupancy in a Large Cancer Center** Oct 2024 – Present
◦ Frames intraday occupancy as a set-function learning problem, where the input is the set of scheduled sessions and the output is the occupancy curve, and proposes a general framework for solving it. Numerical experiments test the framework with machine learning models (Random Forest, XGBoost, neural networks) alongside Gaussian processes, simulation, and time-series approaches.
- **Predictive Scheduling in Service Systems with Simulation and Optimization** Mar 2023 – Present
◦ Study whether the realized execution of an appointment schedule can be predicted before the service day and develop a simulation–optimization framework for feasible schedule prediction under operational uncertainty.

HEALTHCARE & PROFESSIONAL EXPERIENCE

- **Data Scientist Intern** May 2025 – Aug 2025
Dana-Farber Cancer Institute, Boston, MA
◦ Took a patient wait-time prediction model from research prototype to a production machine-learning system used in clinical operations.
◦ Built scalable Databricks pipelines using PySpark, Delta Lake, and MLflow across 1M+ patient records, including automated retraining, drift monitoring, and performance evaluation.
◦ Collaborated with 15+ clinicians, product managers, and engineers to align model outputs with operational workflows and communicate results to clinical leadership.

SELECTED PRESENTATIONS

- **Oct 2025:** “Dynamic Interval Prediction for Patient Wait Times in a Cancer-Care Facility,” INFORMS Annual Meeting, Atlanta, GA.
- **Jul 2024:** “Dynamic Interval Prediction for Patient Wait Times in a Cancer-Care Facility,” INFORMS MSOM Conference, Minneapolis, MN.
- **Aug 2023:** “Physician Rostering Problem with Downstream Capacity Constraints,” INFORMS Healthcare Conference, Toronto, Canada.

TECHNICAL EXPERTISE

- **Programming & Platforms:** Python, SQL, R, MATLAB, VBA; PySpark, Databricks, Snowflake, MLflow, Delta Lake, Gurobi
- **AI & Analytics:** Regression and classification, neural networks, survival analysis, probabilistic forecasting, time-series modeling, NLP/text mining, explainability
- **Operations & Decision Sciences:** Scheduling, predictive analytics, queueing and service systems, mathematical optimization, simulation